Unit 3: Measurement — Predicted Test Questions

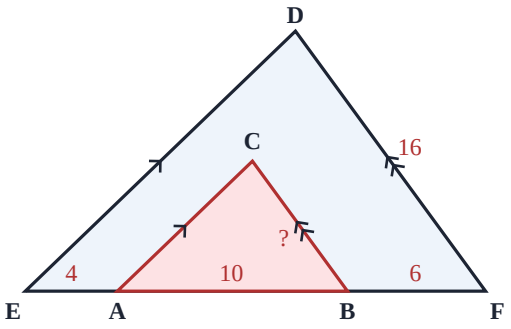
Math 9E · Lessons 3.1–3.4

These 12 questions (3 per lesson) are the ones your teacher is most likely to ask, based on your worksheets: the ★★★/★★★★ practice questions, the proofs done in class and the "Challenge before next class". Each question has a short note explaining why it is likely. Try them as a practice test first; worked answers are at the end.

Questions

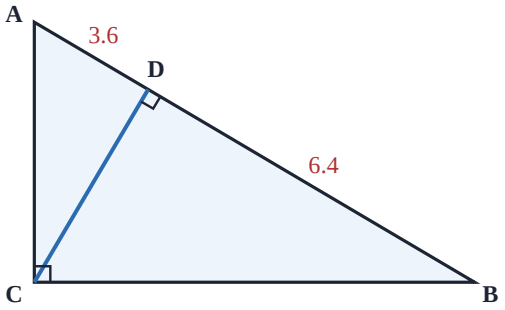
Lesson 3.1 — Similar Triangles

Q1 In ΔEFD, points A and B lie on EF with EA = 4, AB = 10 and BF = 6. Point C is inside the triangle with AC ∥ ED and BC ∥ FD. FD = 16. (a) Prove ΔABC ∼ ΔEFD. (b) Find BC.



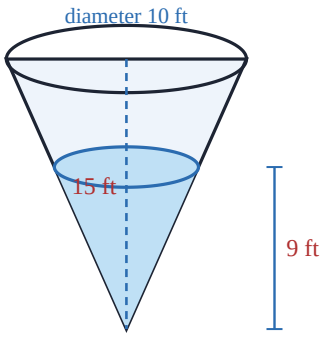
Why: Practice 3.1 #9 asks you to prove two triangles similar and then find x.

Q2 In right ΔABC (right angle at C), CD ⊥ AB. Prove $CD^2 = AD \cdot DB$. Then, if AD = 3.6 and DB = 6.4, find CD, AC and BC.



Why: you were asked in class to prove $CD^2 = AD \times DB$. Expect a proof followed by numbers.

Q3 A conical tank (point down) is 15 ft deep with top diameter 10 ft. It is filled to a depth of 9 ft. Find (a) the area of the water surface and (b) the volume of water.



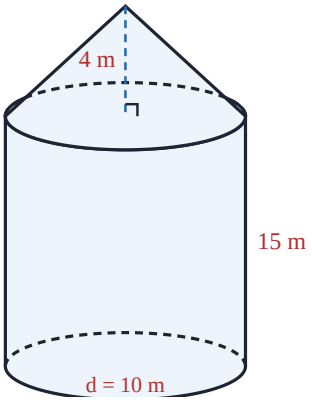
Why: Practice 3.1 #5 (water surface in a conical tank), with volume added.

Lesson 3.2 — Unit Conversions & Surface Area

Q4 A doctor orders 0.6 g of syrup medicine. The label reads 250 mg in 5 mL. How many mL should be given? How many teaspoons is that (1 tsp = 5 mL)?

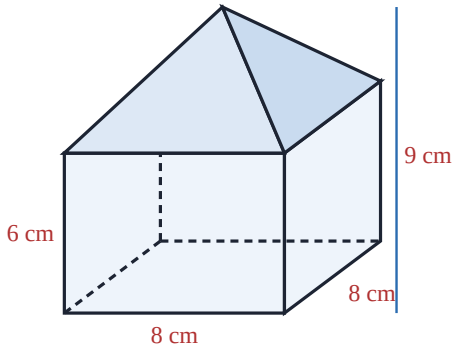
Why: Practice 3.2 #3–#4. The trap is the g → mg conversion.

Q5 A silo is a cylinder with a conical roof. Diameter 10 m, cylinder height 15 m, roof height 4 m. The walls and roof need sealant (not the base). One bucket covers 30 m² and costs $65. Find the area (2 decimal places), number of buckets and total cost.



Why: Practice 3.2 #8 (silo sealant). You must find the slant height with Pythagoras first.

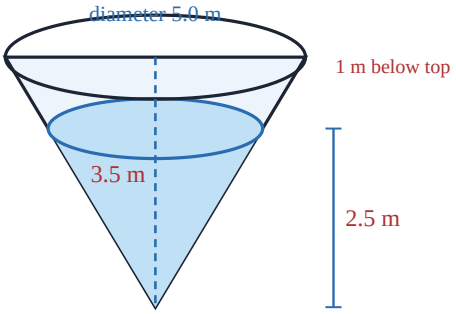
Q6 A square pyramid sits on top of a box with an 8 cm × 8 cm base and walls 6 cm tall. The total height of the shape is 9 cm. Find the total surface area.



Why: Practice 3.2 #11 (pyramid on a box) with new numbers.

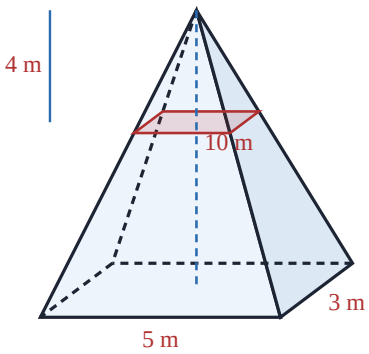
Lesson 3.3 — Volume & Capacity

Q7 An underground tank is a right cone with its apex pointing down. Base diameter 5.0 m, height 3.5 m. (a) Find its capacity in kL to the nearest tenth. (b) How much water is in it when the water level is 1 m below the top? (c) What percentage full is it?



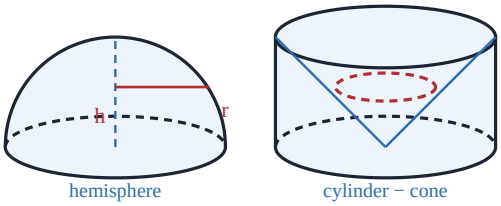
Why: Practice 3.3 #4b (★★★+), the "1 m below the top" twist that needs k^3.

Q8 A right rectangular pyramid has base 5 m × 3 m and height 10 m. A horizontal cut is made 4 m from the apex and the top pyramid is removed. Find the volume of the remaining piece.



Why: Practice 3.3 #9 (★★★+), cut pyramid with different numbers.

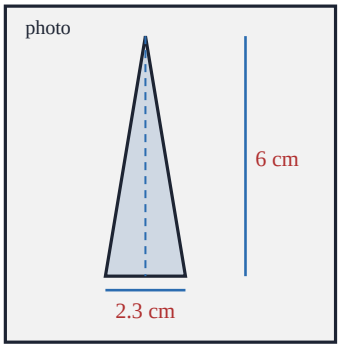
Q9 Derivation. Use Cavalieri's principle with a hemisphere and a cylinder with a cone removed (same radius r and height r) to derive $V_{sphere} = (4/3)\pi r^3$.



Why: Practice 3.3 #13 (★★★★), deriving the sphere formula.

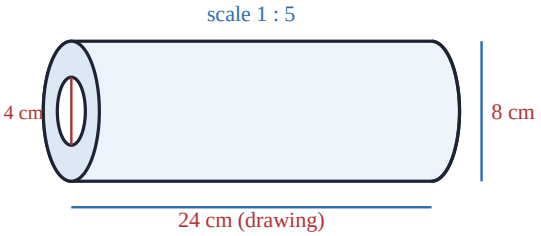
Lesson 3.4 — Scale & Scale Diagrams

Q10 A photo shows a tower. On the photo the tower is 6 cm tall and its base is 2.3 cm wide. The real tower is 324 m tall. (a) Write the scale of the photo as a ratio 1 : n. (b) Find the real width of the base.



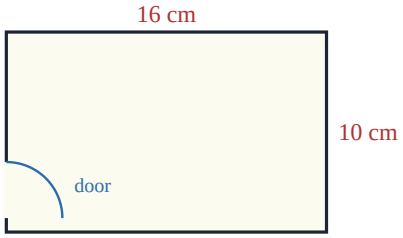
Why: Practice 3.4 #4 (Golden Gate photo) uses this exact two-step pattern.

Q11 A scale diagram of a pipe has scale 1 : 5. On the diagram the pipe is 24 cm long, its outer diameter is 8 cm and the circular opening is 4 cm across. Find (a) the real length, (b) the real diameter of the opening and (c) the real volume of material in the pipe.



Why: Practice 3.4 #7 (★★★) asks exactly these three parts. Part (c) mixes scale with the volume of a cylinder.

Q12 A bathroom is drawn on grid paper where 4 cm represents 1 m. On the plan it is 16 cm × 10 cm. The walls are 3 m tall and have one door of 0.9 m × 2.1 m that is not painted. Tiling costs \$225/m² and painting costs \$16/m². Find the total cost including 12% tax.



4 cm represents 1 m

Why: Practice 3.4 #8 (★★★★ bathroom): scale, area, perimeter and cost.

Answers & Worked Solutions

Lesson 3.1 — Similar Triangles

Q1 (a) AC ∥ ED ⇒ ∠CAB = ∠DEF (corresponding angles). BC ∥ FD ⇒ ∠CBA = ∠DFE. So ΔABC ~ ΔEFD (AA). (b) EF = 4 + 10 + 6 = 20, k = AB/EF = 10/20 = 1/2 ⇒ BC = ½ · FD = **8**.

Q2 Proof: ∠A = ∠DCB (both equal 90° − ∠ACD) and both triangles have a right angle at D ⇒ ΔADC ~ ΔCDB (AA) ⇒ AD/CD = CD/DB ⇒ $CD^2 = AD \cdot DB$ ∎. Numbers: $CD^2 = 3.6 \times 6.4 = 23.04$ ⇒ **CD = 4.8**. AB = 10. $AC^2 = 3.6 \times 10 = 36$ ⇒ **AC = 6**. $BC^2 = 6.4 \times 10 = 64$ ⇒ **BC = 8**.

Q3 r = 5 × 9/15 = 3 ft. (a) $\pi(3^2)$ = **$9\pi \approx 28.27$ ft²**. (b) ⅓π(9)(9) = **$27\pi \approx 84.82$ ft³**.

Lesson 3.2 — Unit Conversions & Surface Area

Q4 0.6 g = 600 mg. 600 × 5/250 = **12 mL = 2.4 tsp**.

Q5 r = 5. Roof slant = $\sqrt{(5^2 + 4^2)} = \sqrt{41} \approx 6.403$. Walls $2\pi(5)(15) = 150\pi$; roof $\pi(5)(6.403) \approx 32.02\pi$. Total ≈ **571.82 m²**. 571.82 ÷ 30 = 19.06 ⇒ **20 buckets ⇒ $1300**.

Q6 Pyramid height = 9 − 6 = 3. Slant = $\sqrt{(3^2 + 4^2)} = 5$. Walls 4(8)(6) = 192; base 64; triangles 4 · ½ · 8 · 5 = 80. Total = **336 cm²**.

Lesson 3.3 — Volume & Capacity

Q7 (a) $V = \frac{1}{3}\pi(2.5^2)(3.5) \approx 22.91$ m³ ≈ **22.9 kL**. (b) Water depth = 2.5 m ⇒ k = 2.5/3.5 = 5/7. $V = 22.91 \times (5/7)^3 \approx$ **8.3 kL**. (c) $(5/7)^3 \approx$ **36.4%**. Being only 1 m below the top still leaves the tank well under half full!

Q8 $V_{big} = \frac{1}{3}(15)(10) = 50$ m³. k = 4/10 = 0.4 ⇒ $V_{small} = 50 \times 0.064 = 3.2$ m³. Remaining = **46.8 m³**.

Q9 Slice both shapes at height h. Hemisphere slice: radius $\sqrt{(r^2 - h^2)}$ (Pythagoras), so area = $\pi(r^2 - h^2)$. Washer: the outer radius is r. The cone (apex at the bottom, height r, radius r) has radius h at height h, so washer area = $\pi r^2 - \pi h^2 = \pi(r^2 - h^2)$. The areas are equal at every height, so by Cavalieri $V_{hemisphere} = V_{cyl} - V_{cone} = \pi r^3 - \frac{1}{3}\pi r^3 = \frac{2}{3}\pi r^3$. Sphere = $2 \times \frac{2}{3}\pi r^3$ = **$(4/3)\pi r^3$** ∎

Lesson 3.4 — Scale & Scale Diagrams

Q10 (a) 6 cm : 32 400 cm ⇒ **1 : 5400**. (b) 2.3 × 5400 = 12 420 cm = **124.2 m**.

Q11 (a) 24 × 5 = **120 cm**. (b) 4 × 5 = **20 cm** (outer diameter 40 cm). (c) $V = \pi(20^2 - 10^2)(120) = 36\,000\pi \approx$ **113 097 cm³ ≈ 113.1 L**.

Q12 Room 4 m × 2.5 m. Floor = 10 m² × $225 = $2250. Walls: perimeter 13 m × 3 m = 39 m² − 1.89 m² (door) = 37.11 m² × $16 = $593.76. Subtotal $2843.76; with tax ×1.12 = **$3185.01**.